COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	17/10/2026	**DATE SUBMITTED:**	20/10/2026
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ACTIVITY NO. 13.1 Parallel Algorithms and Multithreading

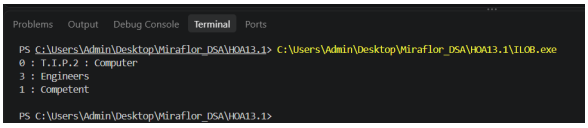
1. PROCEDURE OUTPUT

What procedures did you perform?

Table 13.1
Part 1:

Screenshot	
Analysis	**The program uses one thread to print an integer and a string. The output shows the number 10 and the text T.I.P. The join() function makes the program wait until the thread finishes its task.**

Table 13.2
Part 2:

Screenshot	
Analysis	**The program uses multiple threads to print four different strings. Each thread has its own number and text to display. The join() function makes sure all threads finish before the program ends. And the output will be randomize.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Part A:

1. Write a definition of multithreading and its advantages/disadvantages.

- **Multithreading is when a program uses multiple threads to do different tasks. Its advantage is that it can make a program faster and do many tasks. Its disadvantage is that it can be hard to fix errors and manage the threads.**

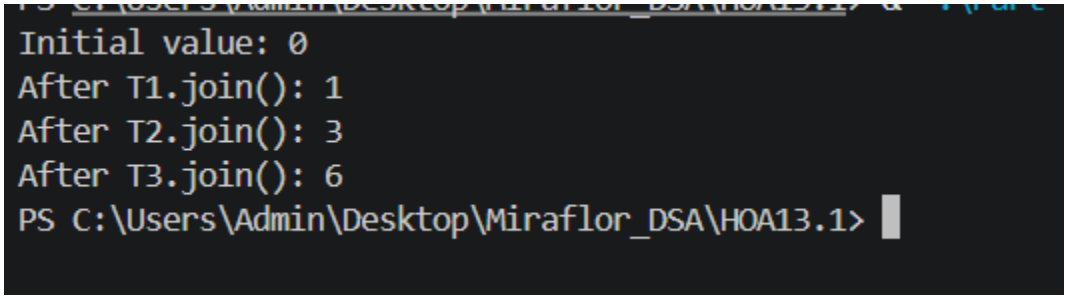
2. Rationalize the use of multithreading by providing at least 3 use-cases.

- **Multithreading can be used in games to handle movements and sounds, in websites to handle many users, and in file processing to help process files faster.**

3. Differentiate between parallelism and concurrency.

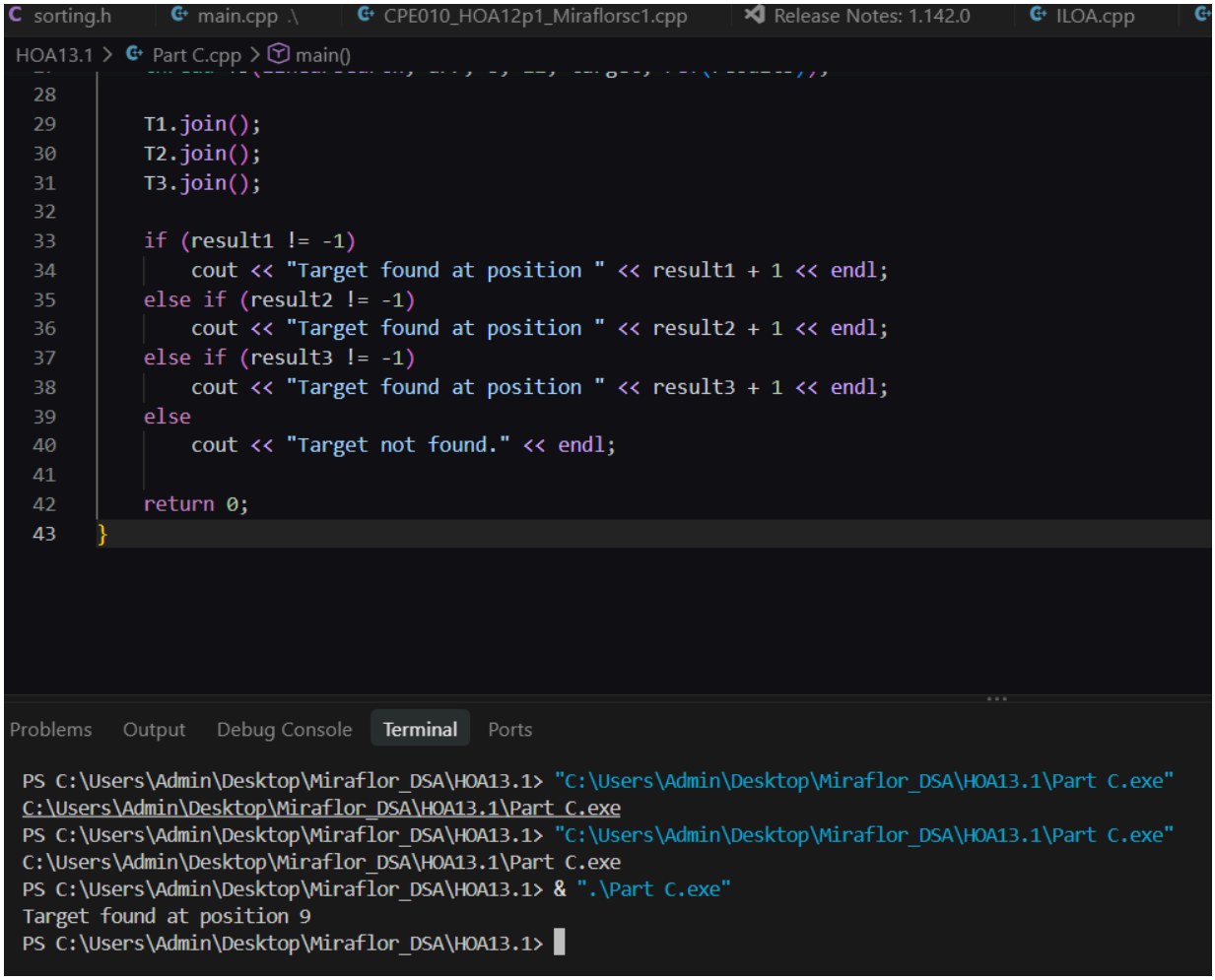
- **Concurrency means handling multiple tasks at the same time by switching between them, while parallelism means actually running multiple tasks at the same time using different processor cores.**

Part B:



Analysis: the program created three threads, and each thread added a number to the global variable. After using join() for each thread, the total became 1, then 3, and finally 6. I learned that join() waits for a thread to finish before displaying the next result.

Part C:



Analysis: In this activity, I used multithreading with the linear search algorithm to find the number 9 in an array. The program divided the array into three parts, and each thread searched its own part. The number 9 was found at position 9. I learned that using multiple threads can help divide a task into smaller parts, but it does not always mean the program will be faster.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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